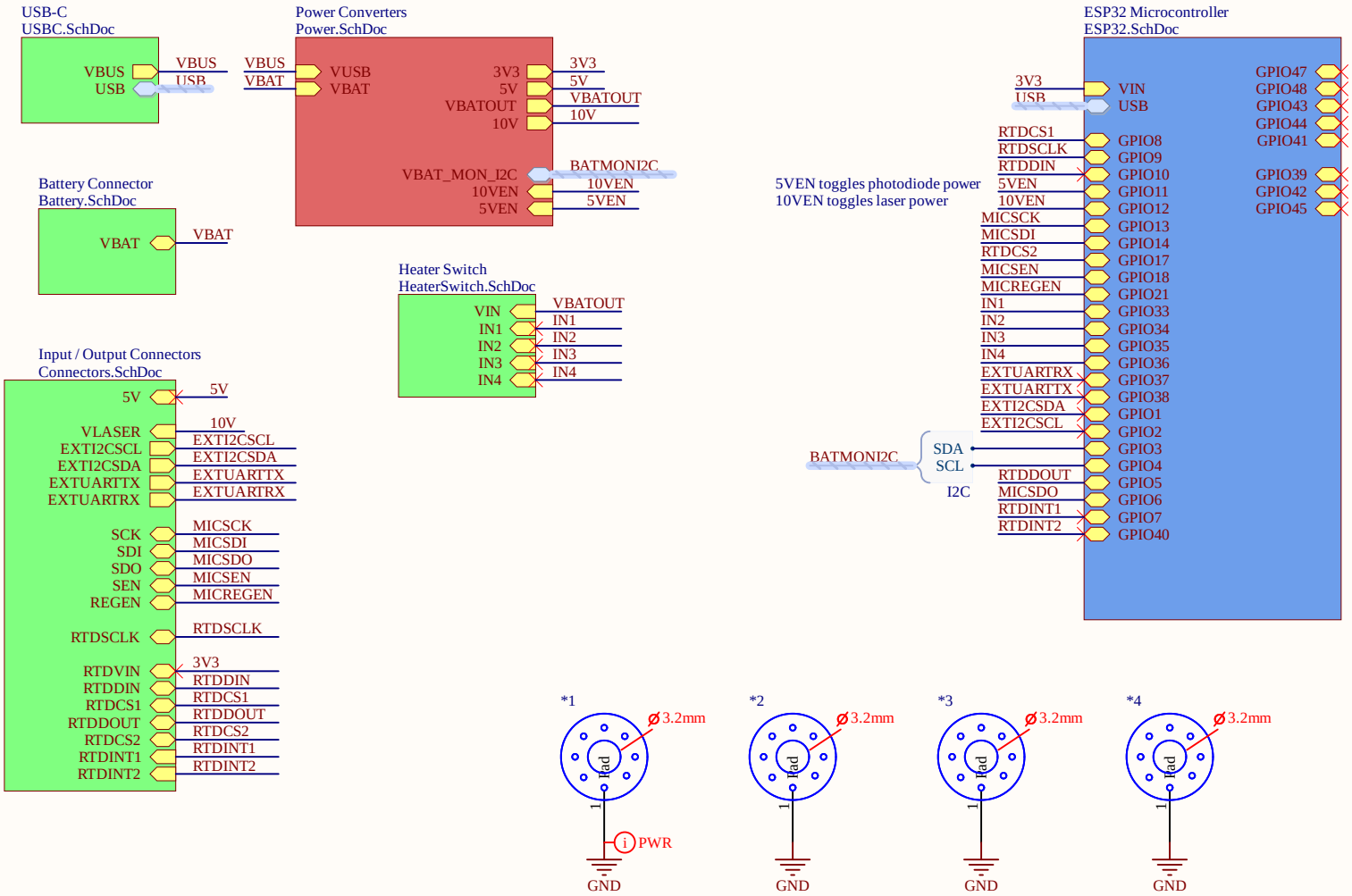


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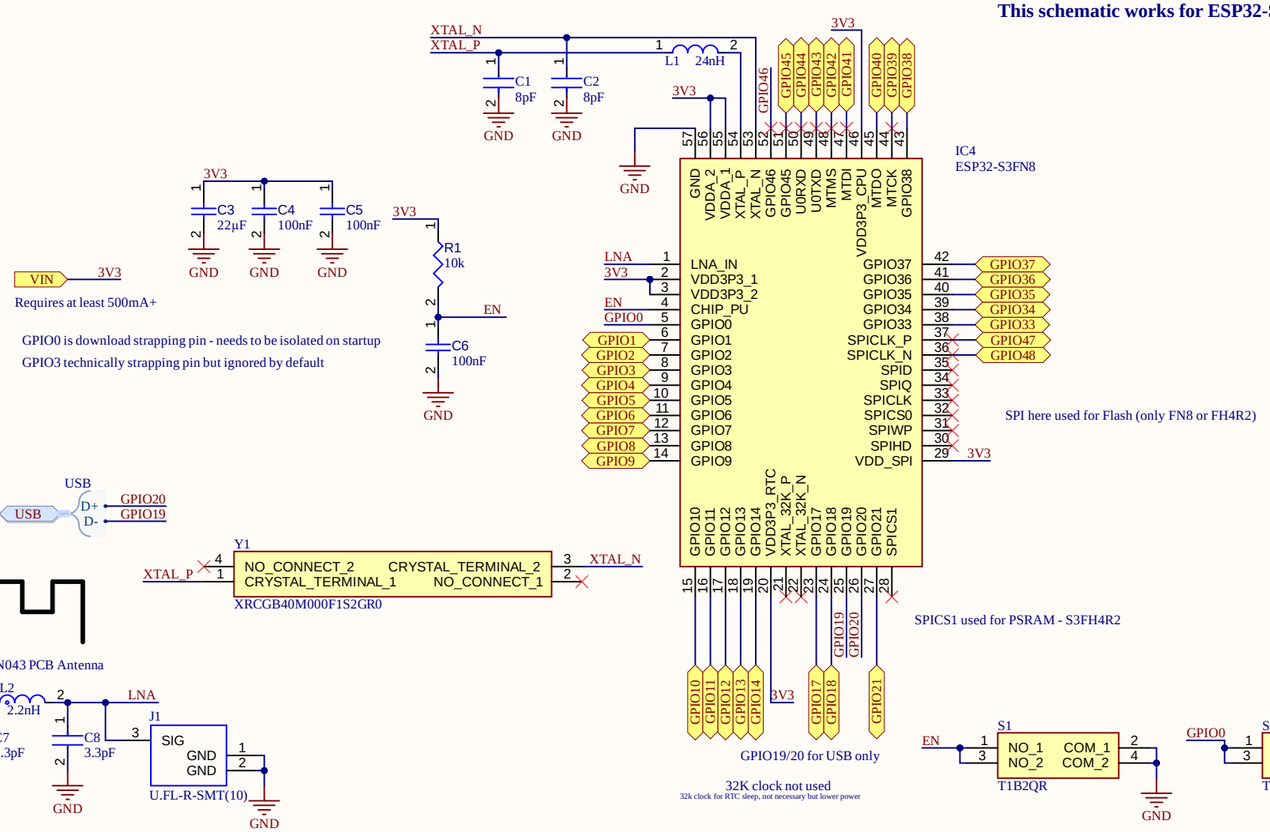
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DWG. NO. | REV | SHT. 2

GPIO46 is a strapping pin, must be low to enter download mode
GPIO45 technically strapping pin but ignored by default

This schematic works for ESP32-S3FN8 or FH4R2



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DWG. NO. | REV | SHT 3

88% Efficiency
Power Loss = 214.25mW
2.5A max output

10V Laser Regulator
LaserRegulator.SchDoc

VIN VOUT → 10V → 10V
EN ← 10VEN

94% Efficiency
Power Loss = 63.25mW
2.5A max output

VUSB Ideal Diode
IdealDiode.SchDoc

VIN VOUT
Power Loss = 0.0041W (4mW)

5V Switching Regulator
ServoRegulator.SchDoc

VIN VOUT → 5V → 5V
EN ← 5VEN

3.3V Switching Regulator
SwitchingRegulator.SchDoc

VIN VOUT → 3V3

98% Efficiency
Power Loss = 38.77mW
2.5A max output

VBAT Ideal Diode
IdealDiode.SchDoc

VIN VOUT
Power Loss = 0.003034W (3mW)

VBATOUT

VUSB → VUSB_Filtered

Use USB if present, otherwise use VBAT

Both resistors should = 8m

R9 16m (1, 2)
R10 16m (1, 2)

VBAT → VBAT_Filtered

VBAT Power Monitor
PowerMonitor.SchDoc

VBAT_Filtered → VBUS
5VPRE → VS
IN+
IN-
ALERT
VBAT_MON_I2C → SDA, SCL (I2C)
SDA
SCL
Uses 19mW
A0 ← 5VPRE
A1 → I2C Address = 0x41
GND

Voltage accuracy = 195uV

8 mohm = 39uA accuracy, 59mW waste (accuracy optimized)

0.5mohm = 625uA accuracy, 3.7mW waste (power optimized)

$$R_{SHUNT} < \frac{V_{SENSE_MAX}}{I_{MAX}}$$

RBAT < 0.16384 / Imax = 0.16384 / 20 = 8.192mohm

If we want more accuracy, choose a value slightly lower than this
If we want lower power loss, choose a significantly lower value
For BAT, we dont care about accuracy (we can calibrate it with the battery removed)

RBAT > 312.5nV / Desired Resolution = 312.5nV / 1mA = 0.3125mohm

Selected resistor = 4mohm
Resolution = 1mA
Power consumption = 4.2 x 0.004 = 0.0168W = 16.8mW

I2C Address = 0x40

are possible and how the data is stored. Below is the Table 4-2 for the INA228. Here we can see that the lowest resolvable shunt voltage neglecting noise corresponds to 312.5nV/LSB or 78.125nV/LSB depending on a ADCRANGE setting found in another register. Also we can see the register LSB does not correspond to the lowest ADC measurement LSB, which is important to be aware of when reading from the device.

Table 4-1. VSHUNT Register Field Descriptions

Bit	Field	Type	Reset	Description
23-4	VSHUNT	R	0h	Differential voltage measured across the shunt output. Two's complement value. Conversion factor: 312.5 nV/LSB when ADCRANGE = 0 78.125 nV/LSB when ADCRANGE = 1
3-0	RESERVED	R	0h	Reserved. Always reads 0.

To find the ideal $R_{SHUNT\ MIN}$ divide the shunt voltage 1 LSB step size by your minimum expected current value.

$$R_{SHUNT\ MIN} \geq \frac{\text{Shunt voltage 1 LSB step Size}}{I_{Sense\ MIN}} \quad (2)$$

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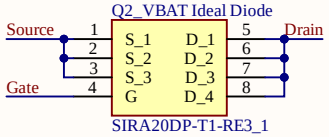
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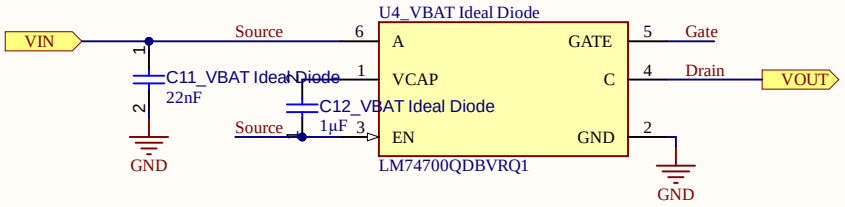
SCALE: | FILE NAME Power.SchDoc | SHEET 3 OF 13

RDSon = 0.00082ohm!!! (0.82mohm)



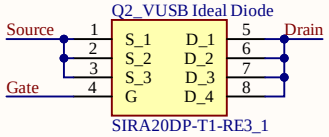
Want an Ideal diode controller with
reverse protection + external MOSFETS
(for lower resistance)
Desired resistance < 10 milliohms

Look at TI ideal diode controller lineup again, specifically LM74704



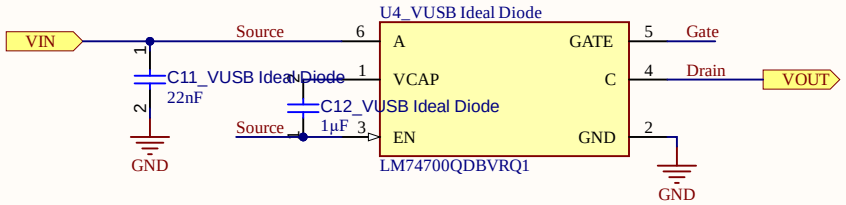
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Git Hash: Version control disabled		**Variant:** [No Variations]	*
File: IdealDiode.SchDoc		**SW Version:**	

RDSon = 0.00082ohm!!! (0.82mohm)

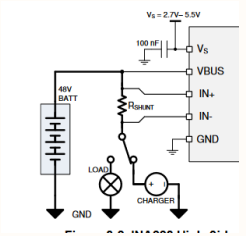


Want an Ideal diode controller with
reverse protection + external MOSFETS
(for lower resistance)
Desired resistance < 10 milliohms

Look at TI ideal diode controller lineup again, specifically LM74704



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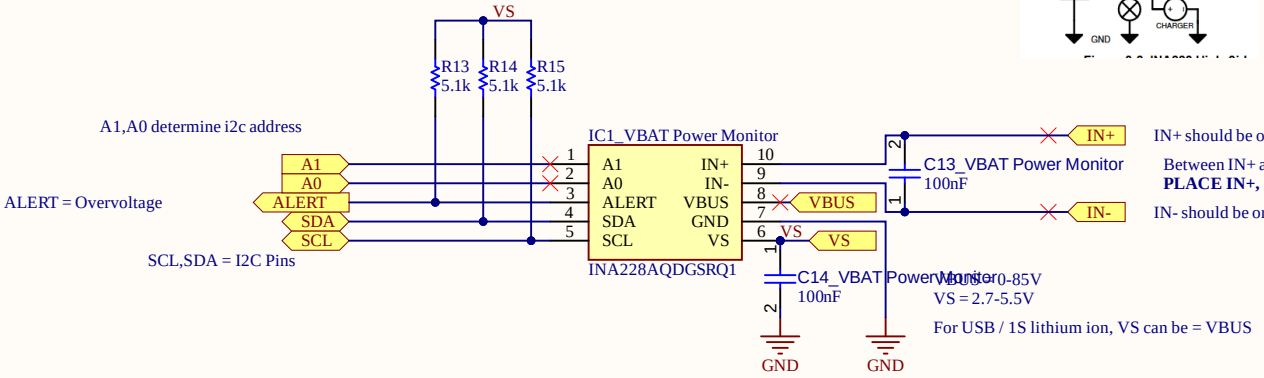
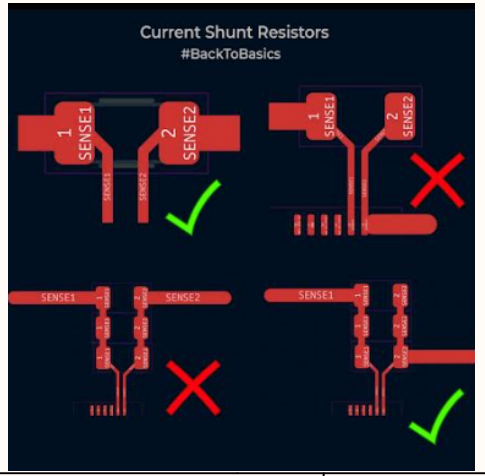


Table 7-2. Address Pins and Secondary Device Addresses

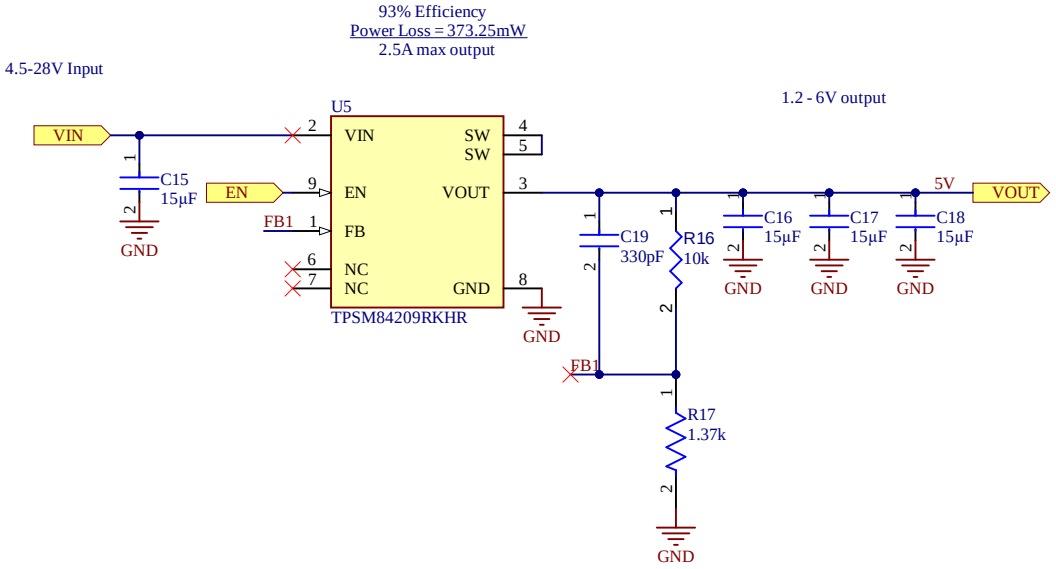
A1	A0	Secondary Device Address
GND	GND	1000000
GND	VS	1000001
GND	SDA	1000010
GND	SCL	1000011
VS	GND	1000100
VS	VS	1000101
VS	SDA	1000110
VS	SCL	1000111
SDA	GND	1001000
SDA	VS	1001001
SDA	SDA	1001010
SDA	SCL	1001011
SCL	GND	1001100
SCL	VS	1001101
SCL	SDA	1001110
SCL	SCL	1001111



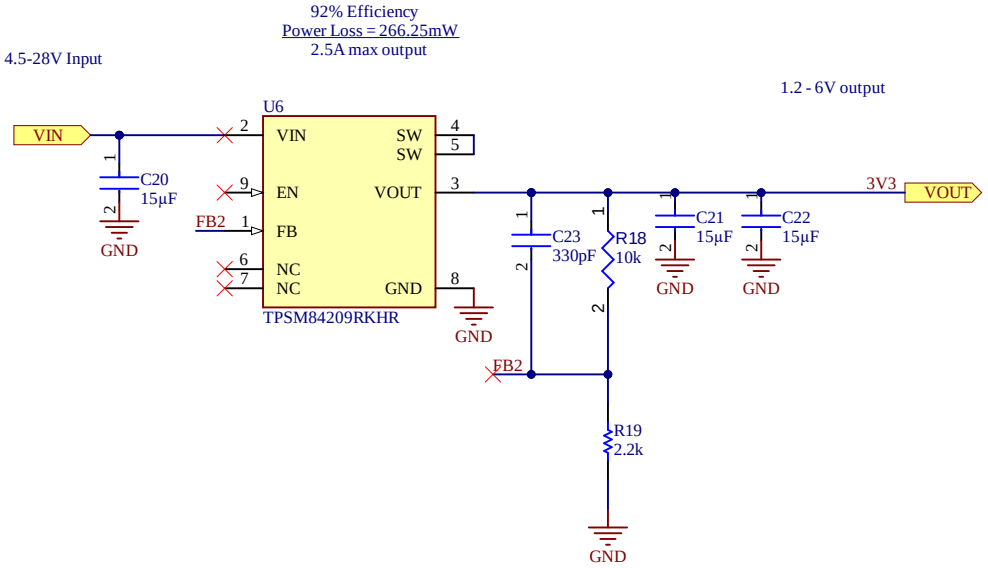
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			Checked: *
Size: A4	**Project:** *	**Rev:** *	**Approved:** *
Date: 8/8/2025 3:51:08 AM	**Sheet** 5 of 13		**Edited:** 7/15/2025
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File: PowerMonitor.SchDoc			**SW Version:**

Space Technologies and Rocketry
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*
*
*

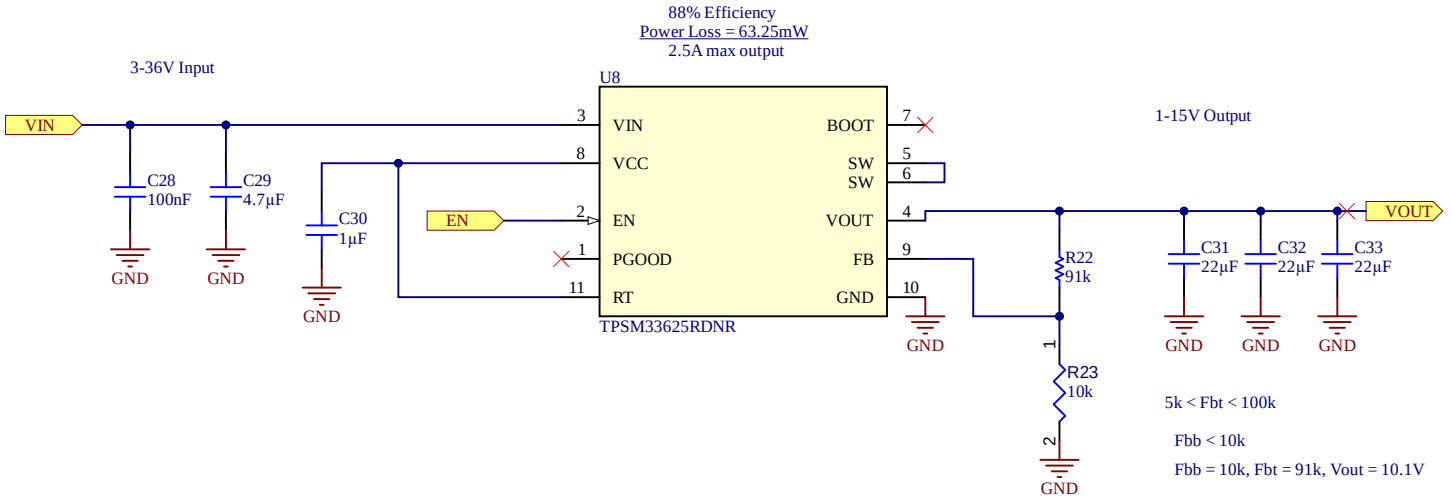
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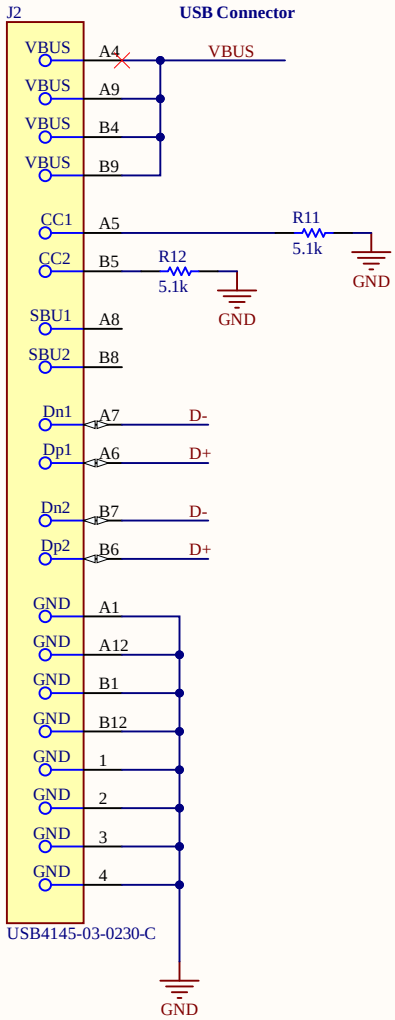
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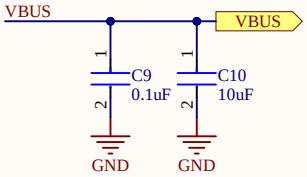


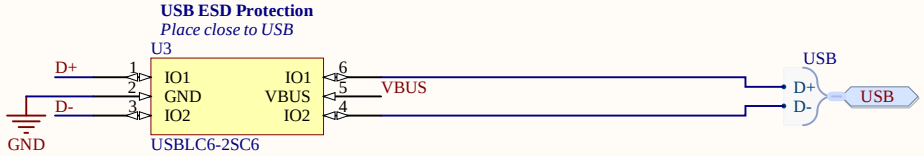
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File: SwitchingRegulator.SchDoc			**SW Version:**	



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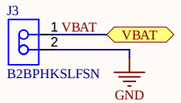


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D1
LD
LDANODE
LDCATHODE
PDANODE
PD
L520P50

Laser Driver

J9
1 VLASER
2
3
4 PDANODE
5 LDANODE
6 LDCATHODE
7
8
9
10
11
12
GND
PPTC121LFBN-RC

Photodiode Input

J4
1 PHOTO_IN
MNT 1
MNT 2
MNT 3
MNT 4
CONBNC001
GND

P1
EXTI2CSCL
EXTI2CSDA
53047-0310
GND

P2
EXTUARTTX
EXTUARTRX
53047-0310
GND

J5
1 5V
2
53047-0210
GND

Microwave Connector

J7B J7A
5V
SCK
SDI
SDO
SEN
REGEN
GND
87831-1241 87831-1241

eeq=sqrt(en^2+(inRf)^2)
LTC6268 eeq = 4.33nv/sqrthz
total input noise
vs 40nv/sqrthz for lmc6482
Photodiode at 3.3V?

C34
100pF
R24
300k
R25
300k
R26
300k
3V3
U9A
LTC6268HS8-10#PBF
PHOTO_IN
PHOTO_PROCESSED
GND

U9B
SHDN
NC
NC
LTC6268HS8-10#PBF

RTD
RTD.SchDoc
RTDVIN
3V3
VIN INT1
INT2
RTDINT1
RTDINT2
RTDDIN
RTDSCLK
RTDCS1
RTDDOUT
RTDCS2
DIN
SCLK
CS1
DOUT
CS2
PHOTO_PROCESSED
RTD1+
RTD1-
RTD2+
RTD2-
RTD3+
RTD3-
RTD4+
RTD4-
RTD5+
RTD5-
RTD6+
RTD6-
RTD7+
RTD7-

7 x 1000 ohm, 2 wire RTDs
J6
RTD1+ 1
RTD2+ 3
RTD3+ 5
RTD4+ 7
RTD5+ 9
RTD6+ 11
RTD7+ 13
2 RTD1-
4 RTD2-
6 RTD3-
8 RTD4-
10 RTD5-
12 RTD6-
14 RTD7-
87831-1411

Pin 1 — Pin 12

Figure 6 LD1100 Pinouts

Pin	Signal	Description
1	+V	Circuit Power, 8 to 12VDC, 250mA, **NOTE**: Internally tied to Laser Diode Anode, the laser anode must be isolated from the power supply COM.
2	COM	Circuit ground
3	V_{REF}	Internal 2.5 V reference
4	PDA	Photodiode Anode
5	LDA	Laser Diode Anode [internally tied to +V (pin 1)]
6	LDC	Laser Diode Cathode
7	RA	When tied to common puts a 100 kΩ in parallel with 249 kΩ internal gain res.
8	RB	When tied to common puts a 33.2 kΩ in parallel with 249 kΩ internal gain res.
9	RC	When tied to common puts a 10 kΩ in parallel with 249 kΩ internal gain res.
10	RD	When tied to common puts a 3.32 kΩ in parallel with 249 kΩ internal gain res.
11	RE	When tied to common puts a 1 kΩ in parallel with 249 kΩ internal gain res.
12	I_{LD}	Laser Diode Current Monitor (10 mV/mA)

IO_5V
C40 10UF
R1 0
R28 0
TP1 DEPOP
Lock Detect
J7
NC
NC
NC
NC
SCK
SDI
SDO
SEN
REG_EN(AD4)
SSW-106-01-T-D
Header to USB Board
TP4 DEPOP
R40 20K
R41 20K

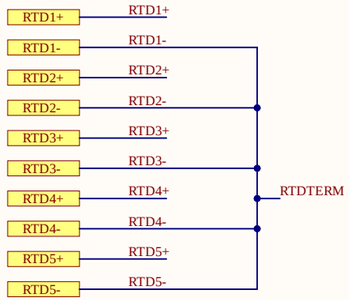
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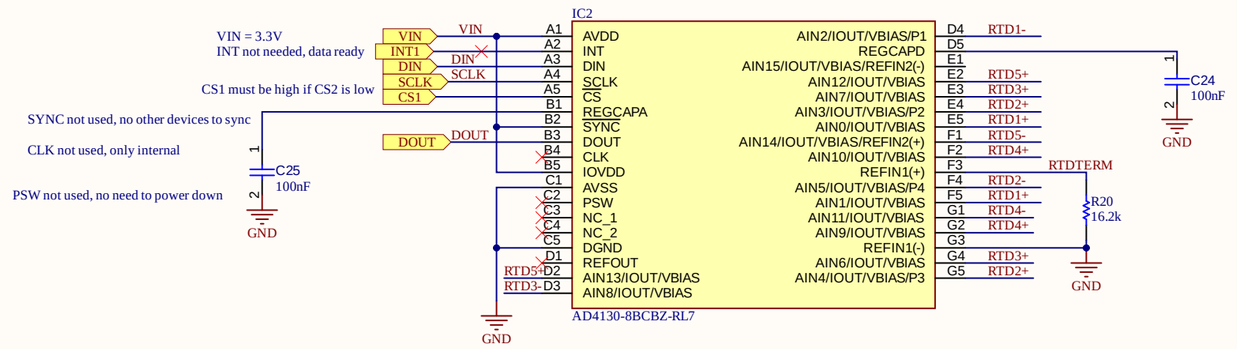
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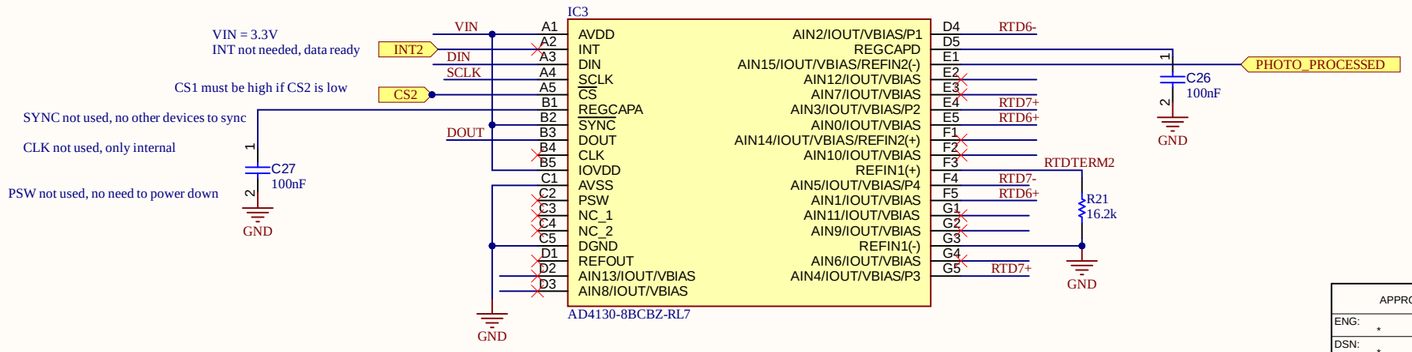
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Using IEXC = 200uA



Using IEXC = 200uA

We have 11 free adc inputs to use for anything

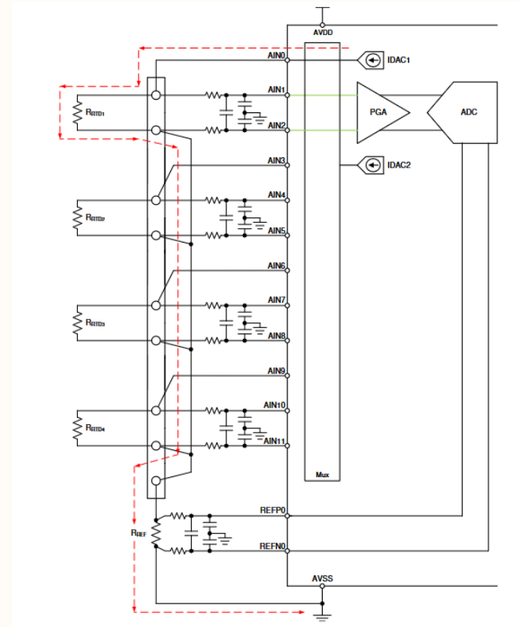
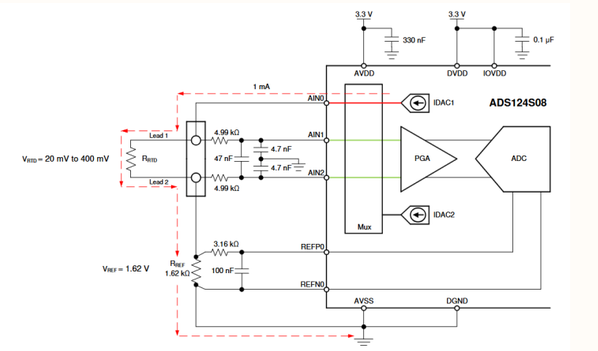


Figure 2-9. Multiple Two-Wire RTDs Measurement Circuit



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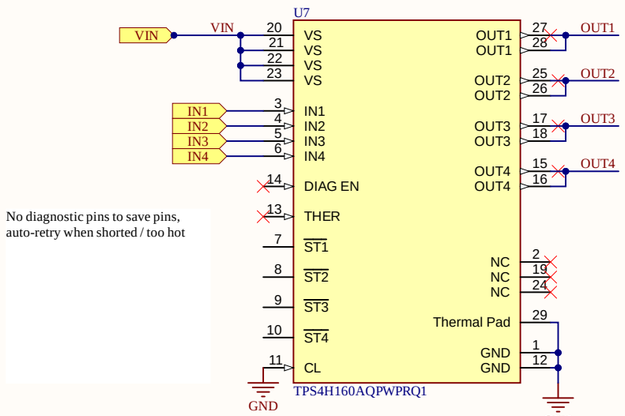
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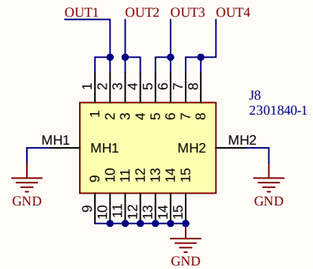
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ASSY DWG:	
FAB DWG:	
PCB DWG:	

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SCALE:	FILE NAME HeaterSwitch.SchDoc	SHEET 13 OF 13	